SELF-X

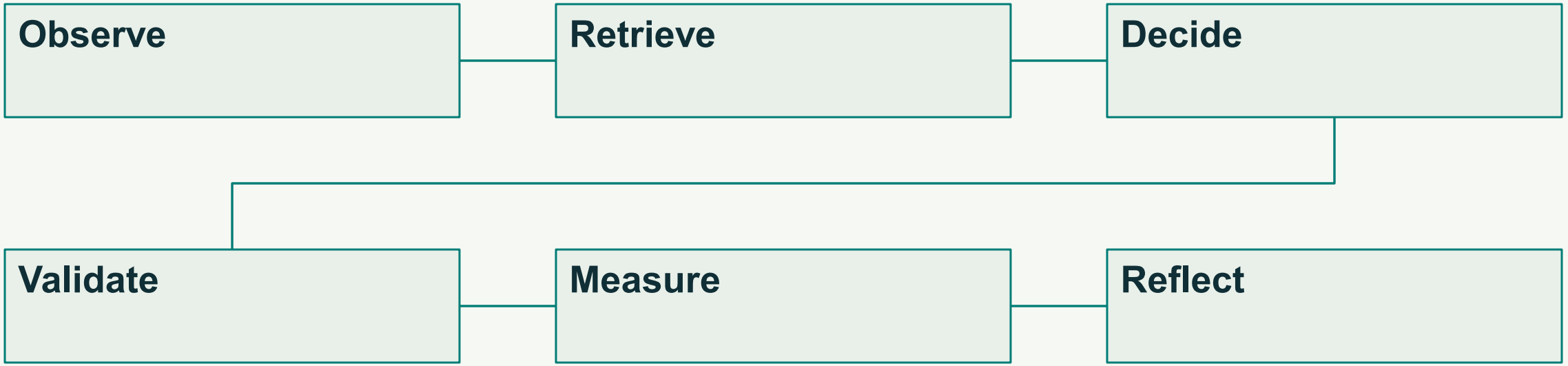
Adaptive episodic memory and lower API control delay

Generated traffic · Paired benchmarks · Real provider requests

Measured revision 3 · 20 October 2026

Project goal and control architecture

Reuse earlier local experience to choose a bounded flow intervention



Source: saved v3 CSV, JSON and provider audits; network values are simulated

Memory retrieval changes

Component	Previous A3	Current A4
State match	Normalized cosine	Magnitude and capacity margins
Action evidence	Optimistic local reward	Separate value per action
Storage	30 balanced episodes	96 diverse episodes
Support	Similar positive episodes	At least 3 matched observations
Selection	Feasible historical preference	Conservative local evidence gate

Current parameters are rebuilt and independently validated for every action.

A4 learns only from completed earlier episodes

API work and local reuse

Legacy: full decision call, then a second API reflection call.

Compact: one short action choice; local parameters and reflection.

Hybrid: exact-context cache, healthy bypass and supported memory.

Pacing and HTTP/service time are recorded separately.

Cache hits and local bypasses are never counted as new API decisions

Generated traffic and primary benchmark coverage

86,400

measured primary control windows

8 scenarios × 10 held-out seeds × 120 windows × 9 controllers

10 Mbps per path · 0.5 Mbit buffer · 5-second action windows

9,600 generated aggregate flow records exported as CSV

Finite-buffer two-path fluid model; no packet capture

Experimental comparisons and evidence boundaries

Study	Sample	Interpretation
Primary held-out suite	86,400 windows; 10 seeds	Ordinary workload comparison
Near-capacity diagnostic	19,200 windows; 10 seeds	Post-hoc mechanism stress test
Fresh Gemini pilot	72 cycles; 2 seeds	System latency and integration
Free model sweep	8/17 valid checks	Availability, not control quality

H1 isolates the base guard. O1 has privileged simulator access.

Every primary controller receives the same demand for each scenario and seed

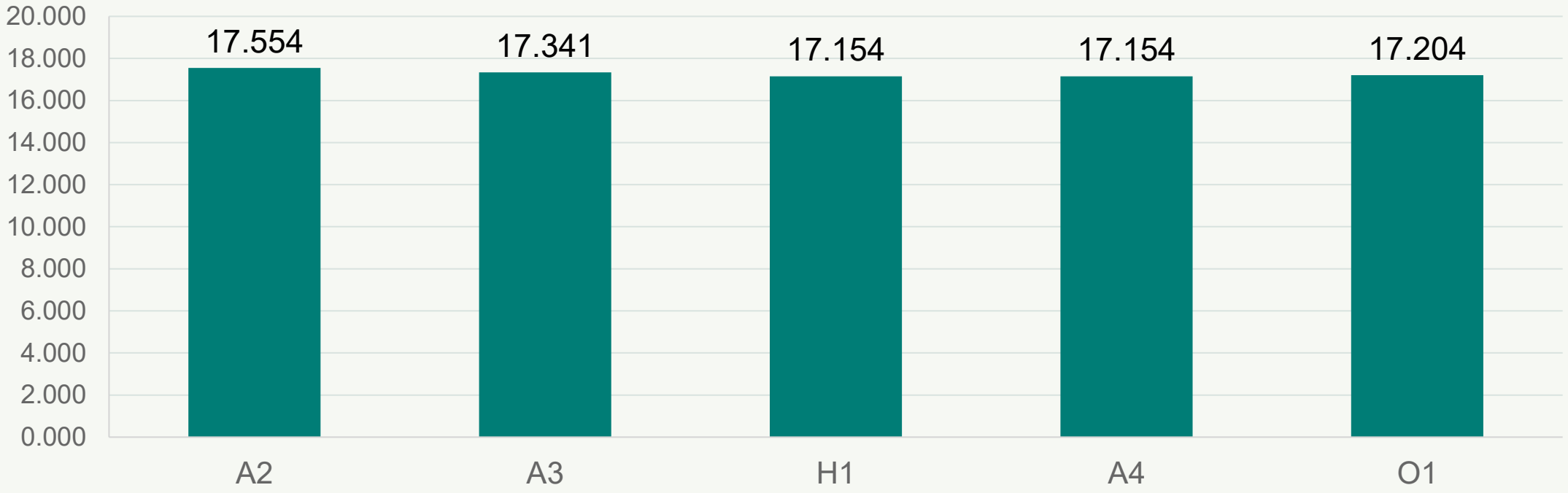
Primary scenario delay results

Scenario	A0	B1	B2	A1	A2	H1	A3	O1	A4
Mixed	30.42	27.58	28.97	24.27	25.10	24.49	24.77	24.39	24.49
Flash	26.58	10.38	18.34	9.83	11.75	9.91	10.98	9.83	9.91
Alt busy	22.36	22.36	22.36	12.95	12.95	13.15	13.15	13.34	13.15
Shift	27.03	15.10	20.52	8.53	8.53	8.59	8.59	9.01	8.59
Overload	35.79	35.79	35.79	35.79	35.79	35.79	35.79	35.79	35.79
Light	8.00	8.00	8.00	8.00	8.00	8.00	8.00	8.00	8.00
Recurring	30.30	22.70	26.42	17.89	17.89	17.89	17.89	17.79	17.89
Volatile	27.54	22.97	25.41	19.22	20.42	19.40	19.55	19.48	19.40

A4 matches H1; it removes A3 regressions on mixed, flash and volatile traffic.

Mean modeled delay in milliseconds; 1,200 windows per cell

Overall primary modeled delay



A4: 17.154 ms. Previous A3: 17.341 ms.
H1 has the same primary network outcome as A4.

Equal weighting across eight primary scenarios and ten paired seeds

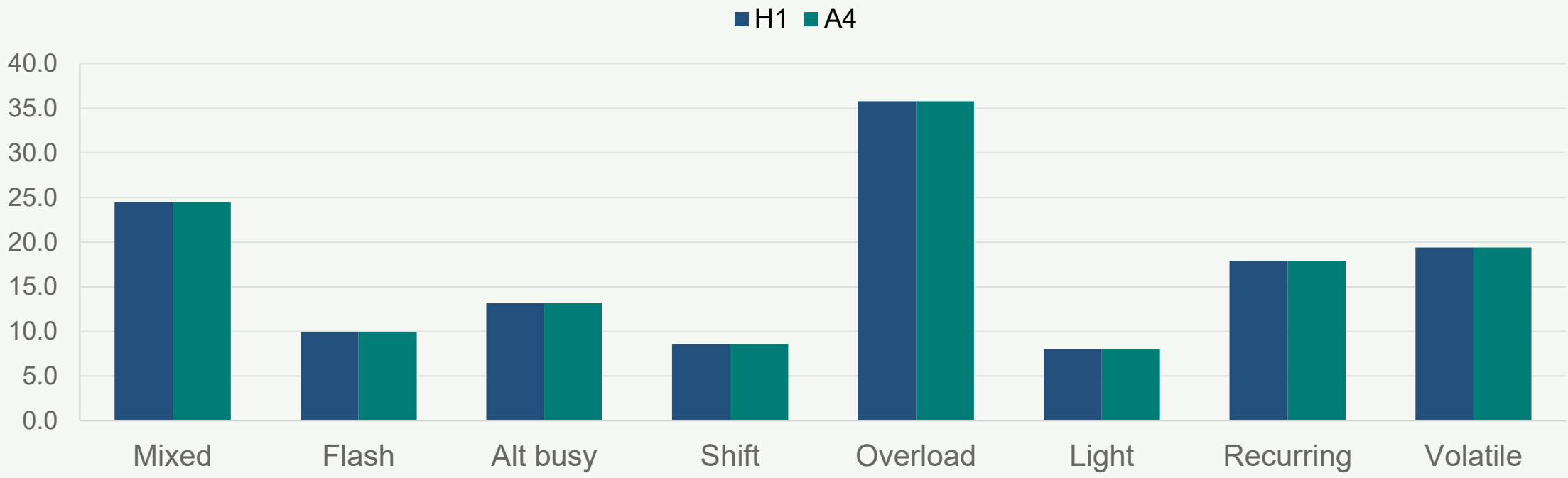
Paired adaptive memory effects

Scenario	Delay difference ms	Paired 95% interval
Mixed	-0.280	[-0.511, -0.106]
Flash	-1.072	[-2.231, 0.000]
Alt busy	0.000	[0.000, 0.000]
Shift	0.000	[0.000, 0.000]
Overload	0.000	[0.000, 0.000]
Light	0.000	[0.000, 0.000]
Recurring	0.000	[0.000, 0.000]
Volatile	-0.145	[-0.319, -0.005]

Exploratory comparisons; intervals do not correct for multiple testing.

A4 minus A3; negative delay differences favor A4

The guard-only memory ablation



A4 and H1 chose the same installed actions in all 9,600 primary windows.

The primary experiment does not show an additional gain over H1

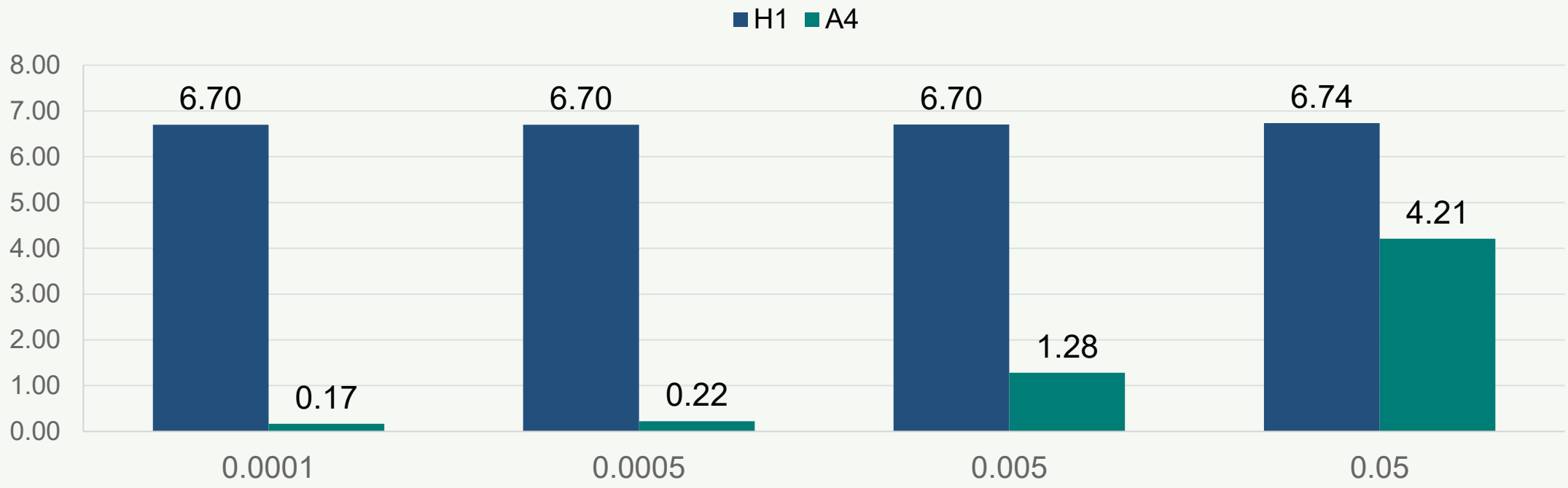
Delay goodput loss and feasibility

ID	Delay ms	Goodput Mbps	Total loss %	Vetoes
A0	26.003	12.904	8.162	0
B1	20.611	13.058	6.986	0
B2	23.227	12.983	7.561	2585
A1	17.062	12.902	7.913	0
A2	17.554	12.848	8.349	347
H1	17.154	12.990	7.382	0
A3	17.341	12.974	7.510	0
O1	17.204	13.001	7.322	0
A4	17.154	12.990	7.382	0

A4 has zero vetoes. Feasibility and performance are separate outcomes.

Loss includes deliberate rate-meter drops; all values are simulated

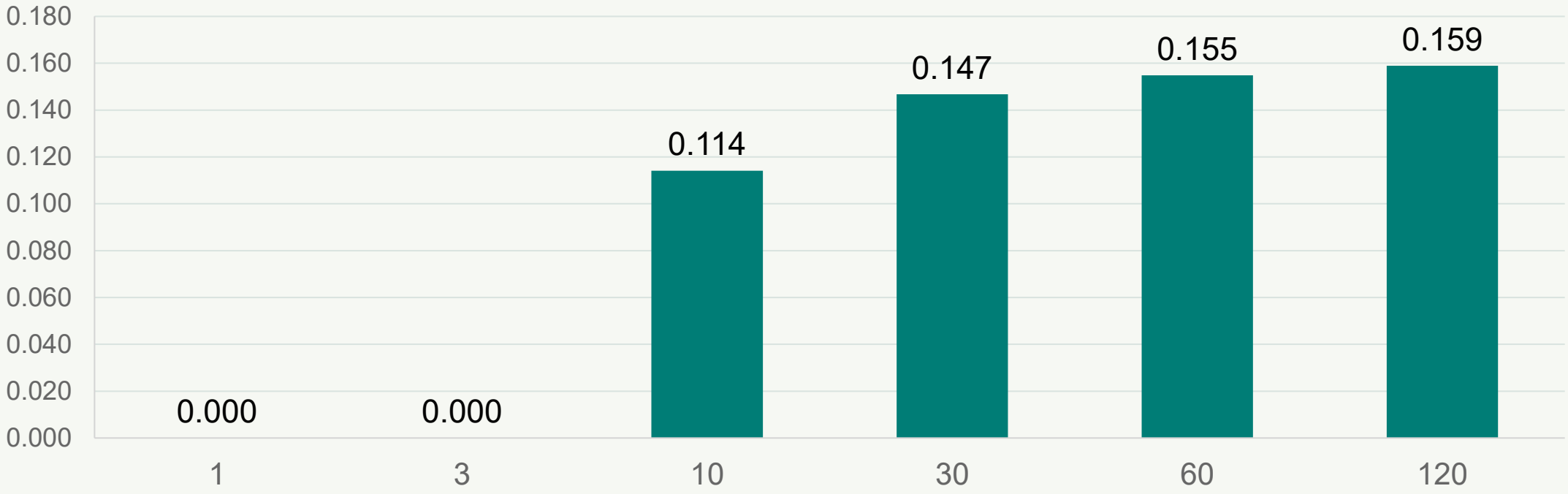
Near-capacity memory sensitivity



Total loss (%) versus primary excess above 10 Mbps.
A4 retains more traffic while allowing higher queue delay.

Constructed post-hoc diagnostic; excluded from the primary average

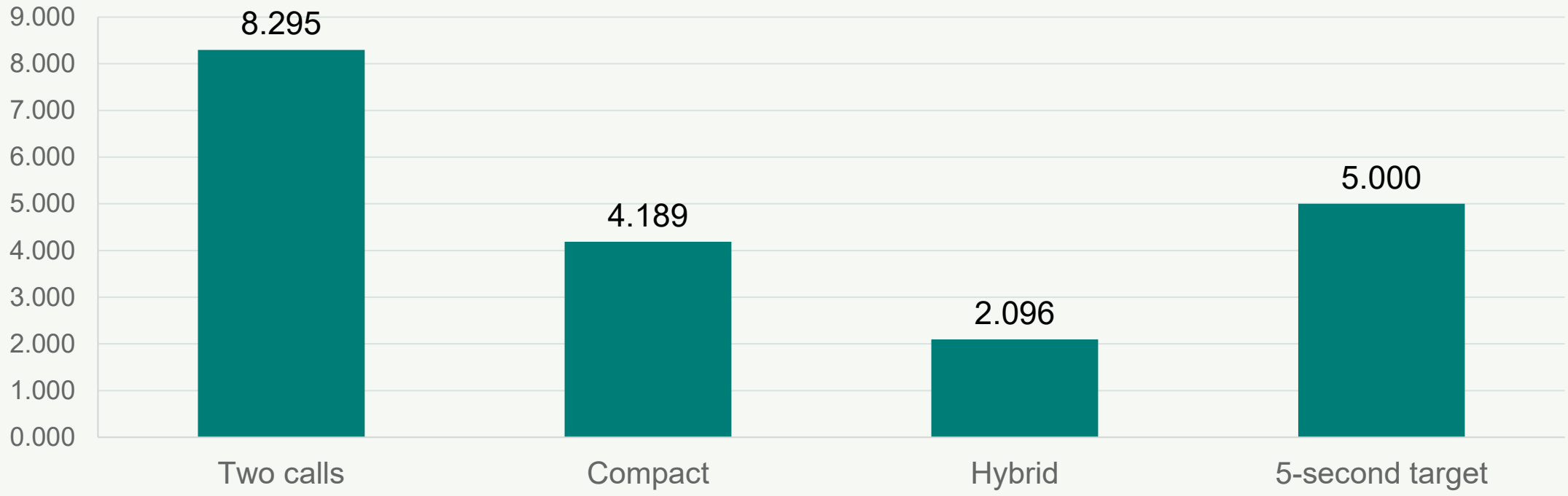
Learning after harmful throttling



Bars: A4 minus H1 cumulative reward; positive favors A4.
Window 120 raw means: A4 -0.004; H1 -0.163.

0.0001 Mbps excess diagnostic; all cold-start windows retained

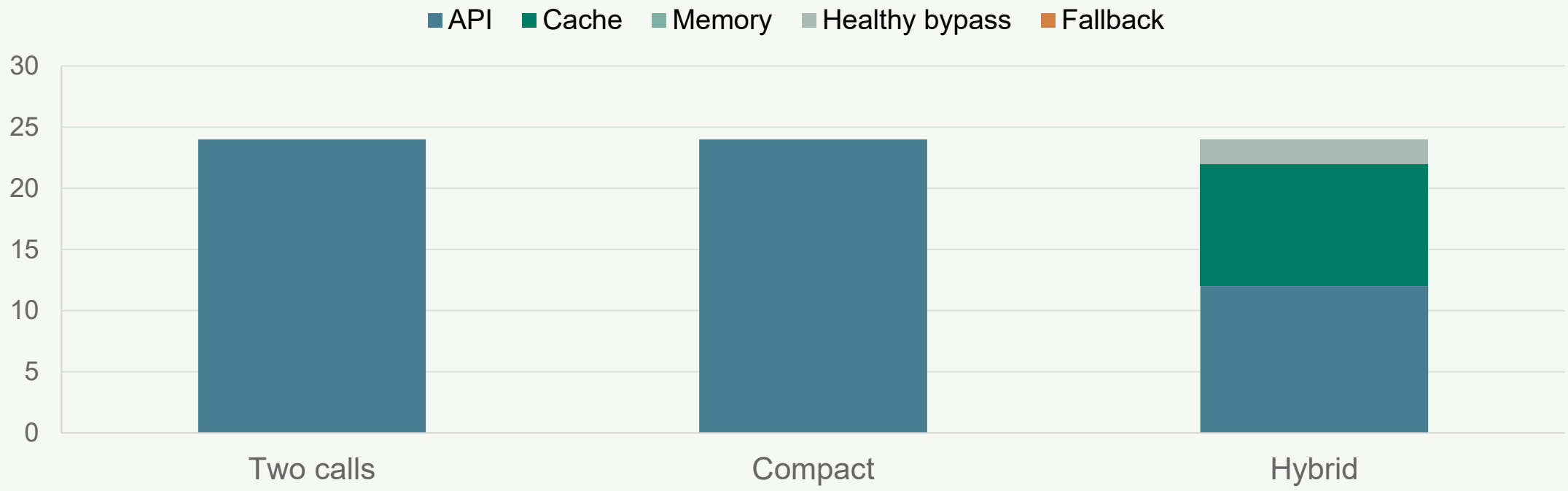
Measured full-cycle Gemini latency



Compact mean reduction: 49.5%. Hybrid: 74.7%.
This measures the whole system, including fewer requests and local reuse.

Two fresh seeds; 24 cycles per mode; pacing and reflection included

Provider calls and decision origins



Actual provider requests: 48 → 24 → 12.
Local measured reflection makes no provider request.

Fresh API responses remain separate from local cache and bypass decisions

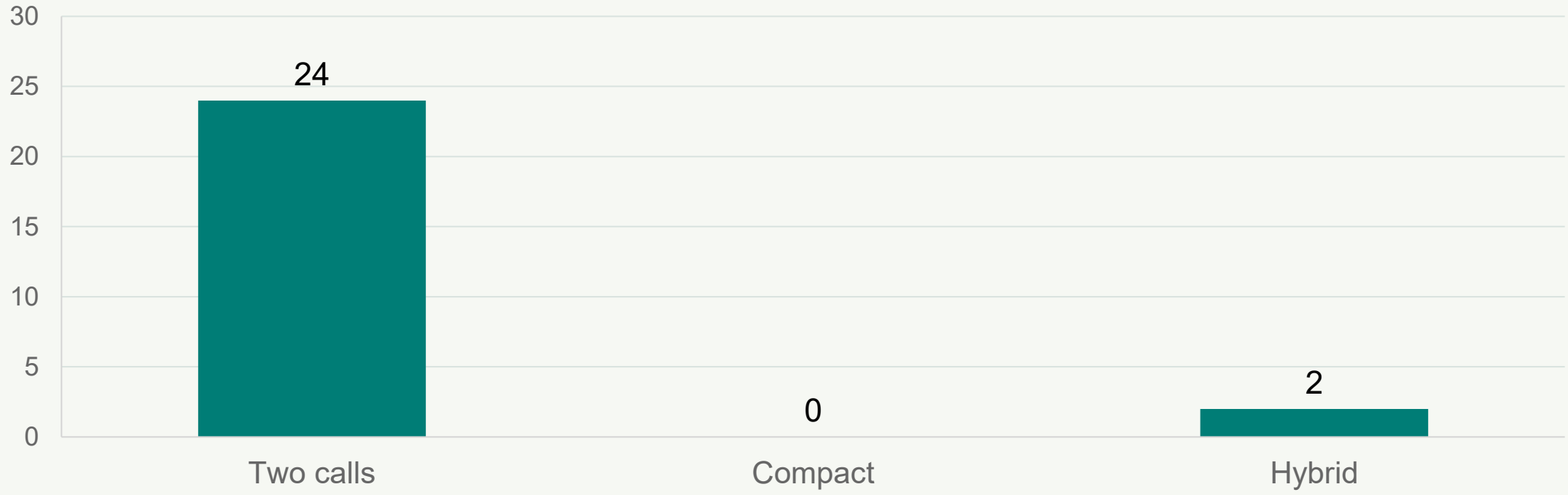
Live pilot network outcomes

Mode	Delay ms	Goodput Mbps	Total loss %	Reward
Two calls	22.760	12.720	6.986	0.111
Compact	17.915	12.637	7.470	0.137
Hybrid	17.915	12.637	7.470	0.137

Hybrid loss minus legacy: 0.484 percentage points.
A latency reduction does not establish a uniform network-quality gain.

Faster execution must be assessed together with action quality

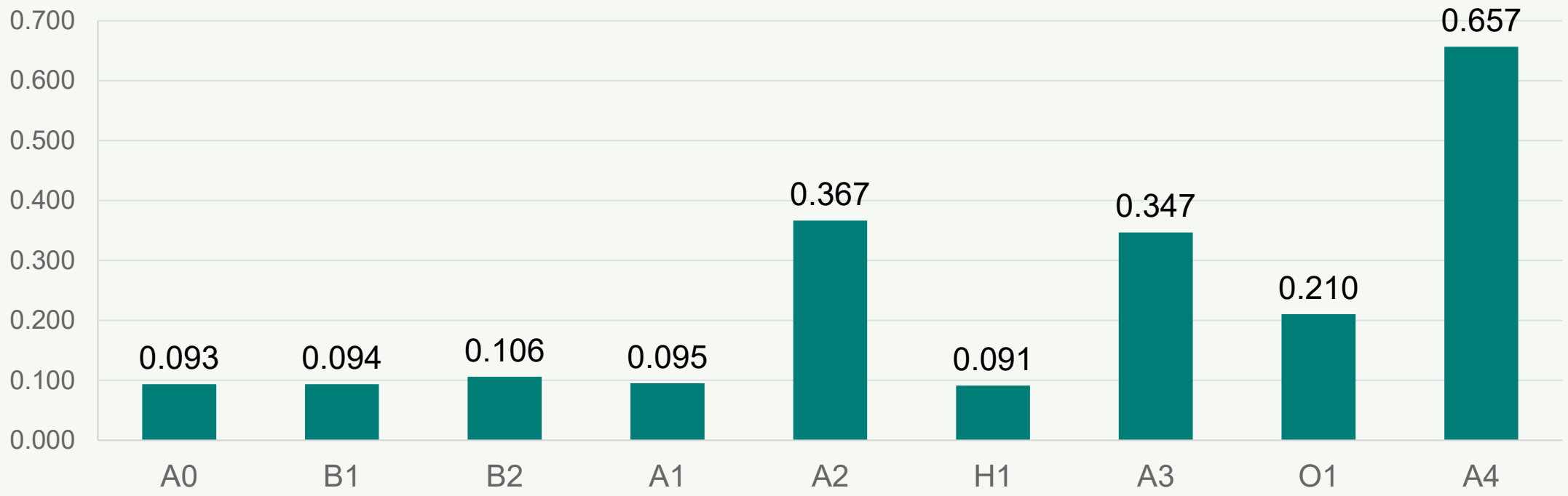
Five-second deadline misses



P95 full cycle: Two calls 8.83 s · Compact 4.56 s · Hybrid 5.71 s
The pilot does not establish a production deadline guarantee.

Observed tail latency remains a practical limitation

Local memory computation overhead



Primary suite wall time: 39.92 seconds.
A4 mean: 0.657 ms; P95: 1.310 ms.

Measured in .venv; excludes the final batched memory flush

Validation and remaining limits

47 passed, 4 subtests passed in 4.21s in the isolated virtual environment.

8/17 compact free-model schema checks returned valid actions.

Primary memory matches H1; targeted benefits have a delay tradeoff.

Network metrics are simulated; Linux packet deployment remains unvalidated.

Reproduce and inspect the results

```
.\run.ps1 all
.\run.ps1 latency
.\run.ps1 dashboard -Port 8766
```

output/benchmarks/v3 → CSV, traffic, API audits and 15 PNG/SVG figures

Editable report and presentation · Report, slide and graph PDFs

Run every command from the selfx folder